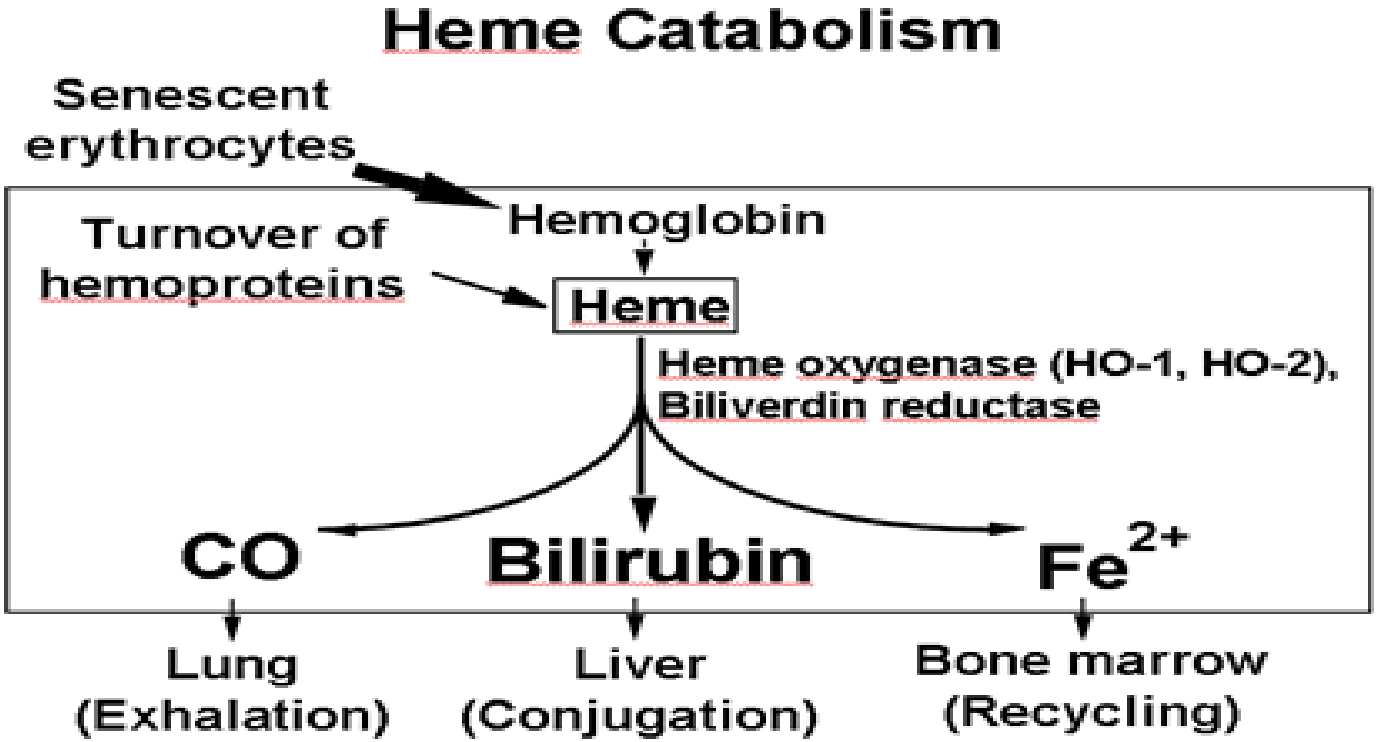
HEMOLYSIS AND BILIRUBIN METABOLISM

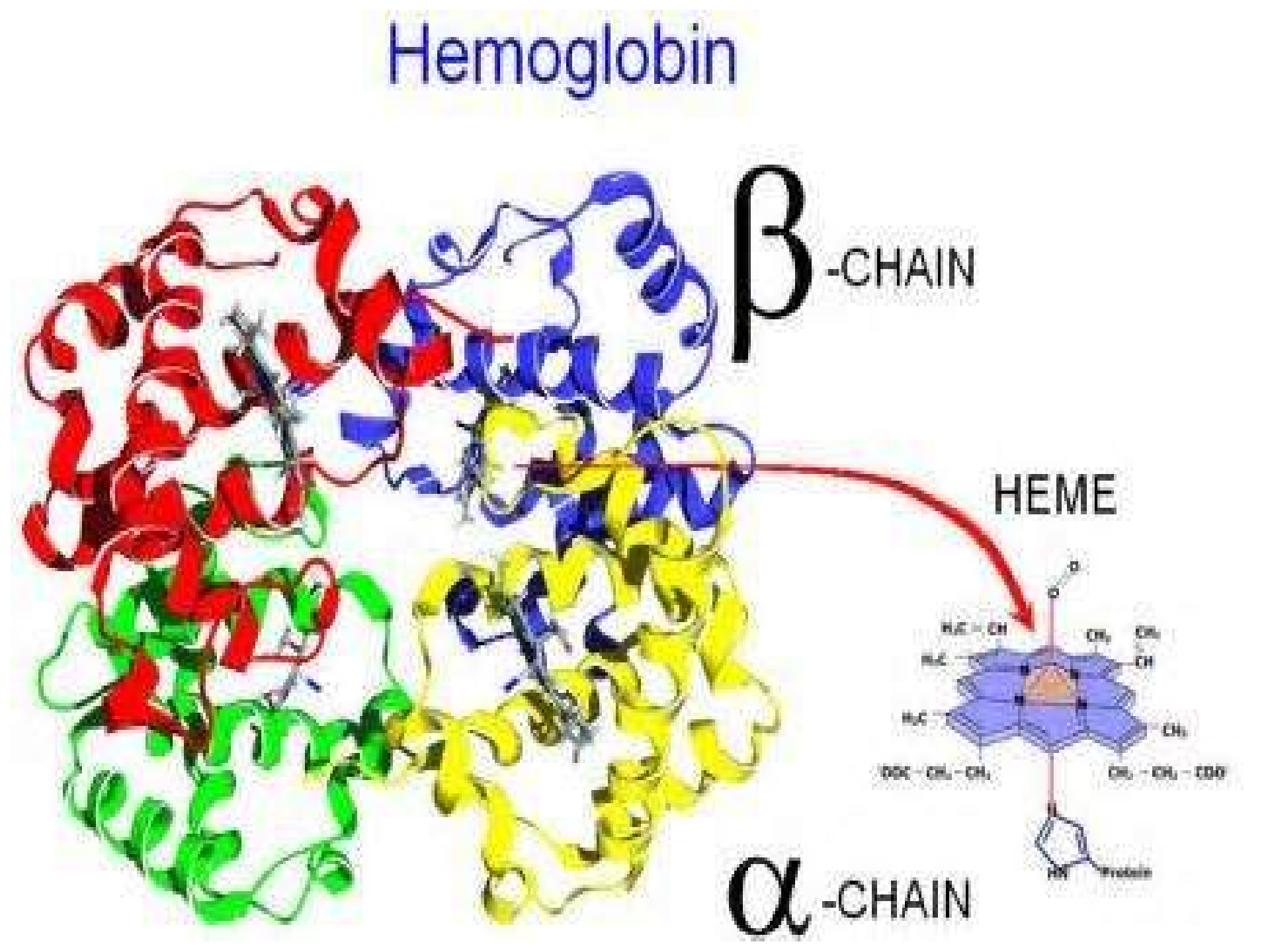


Lecture Objectives

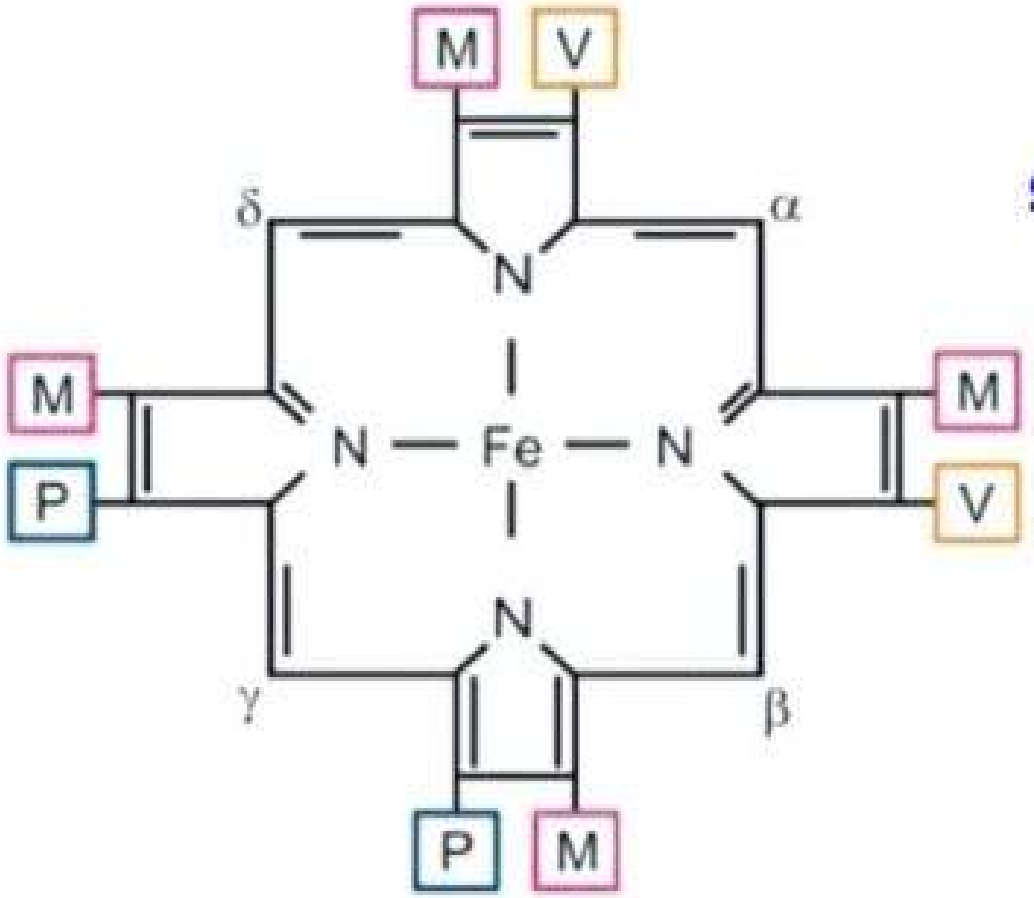
At the end of this session students will be able to understand the following;

1. Degradation of Heme.
2. Formation of Bile pigments.
3. Types of Bile pigments.
4. Transport and excretion of bile pigments.

Hemoglobin



Structure of Heme



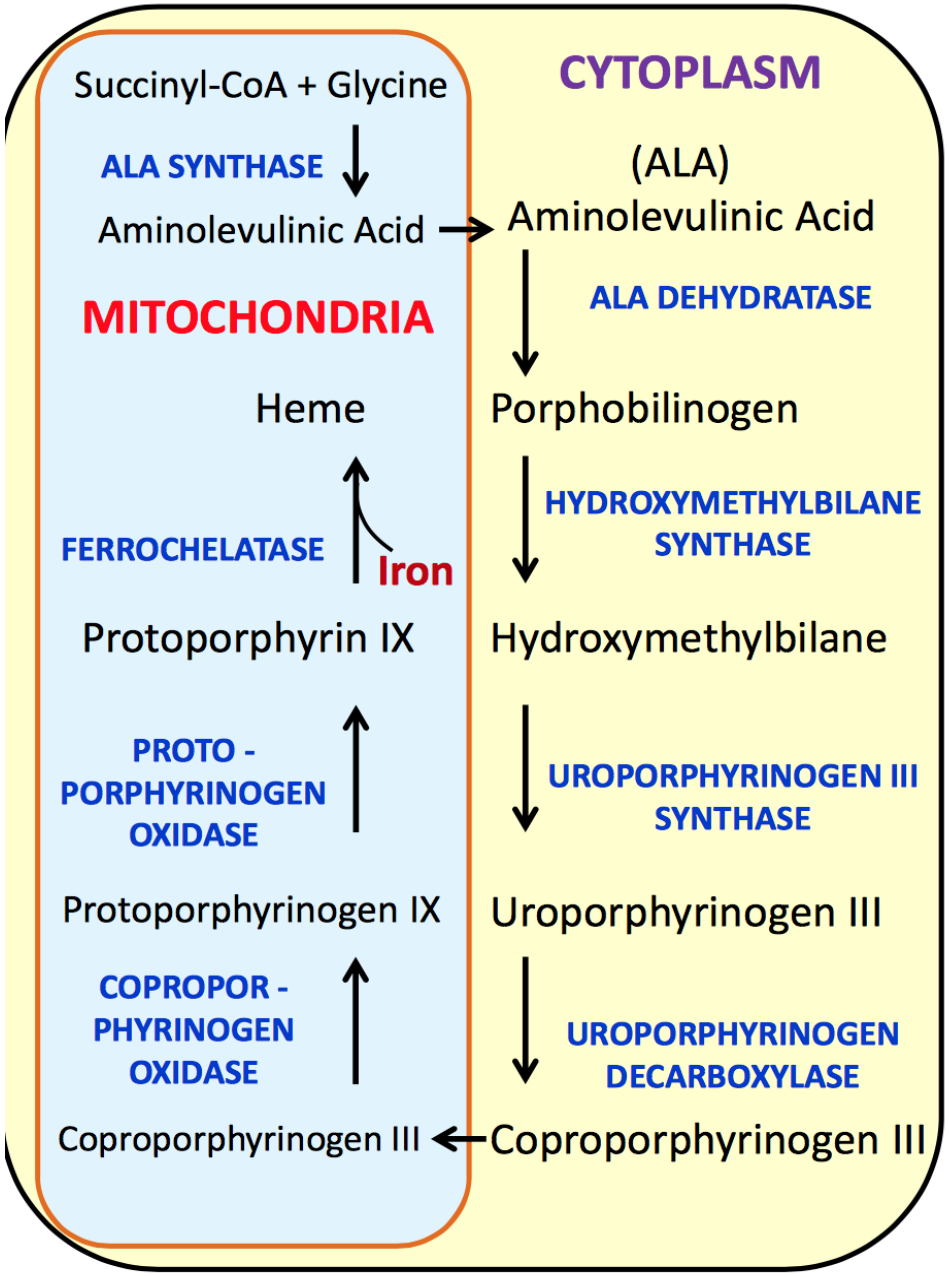
Structure of heme

M = Methyl

V = Vinyl

P = Propionyl

Heme biosynthesis



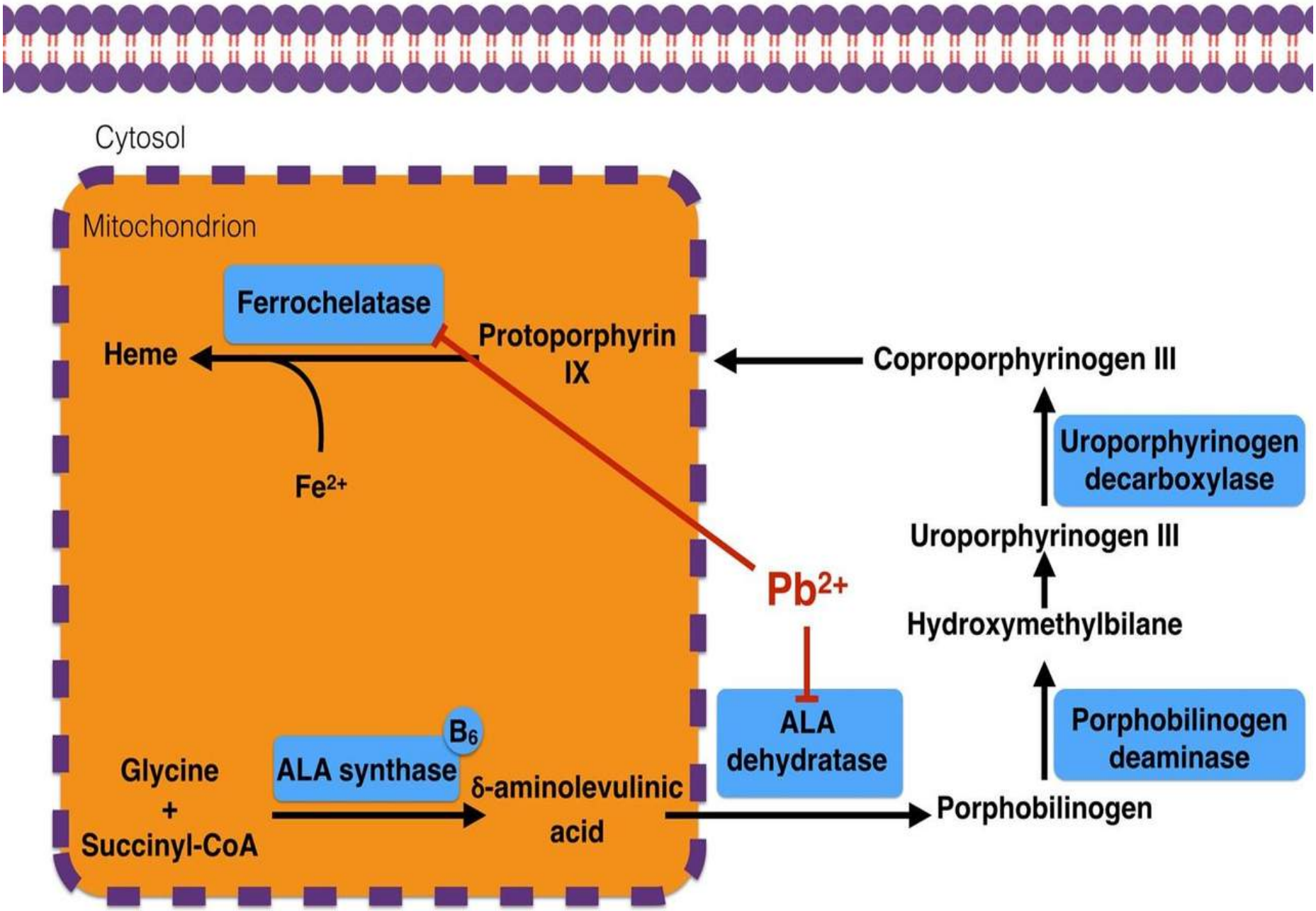
Major **sites** of synthesis in the body:

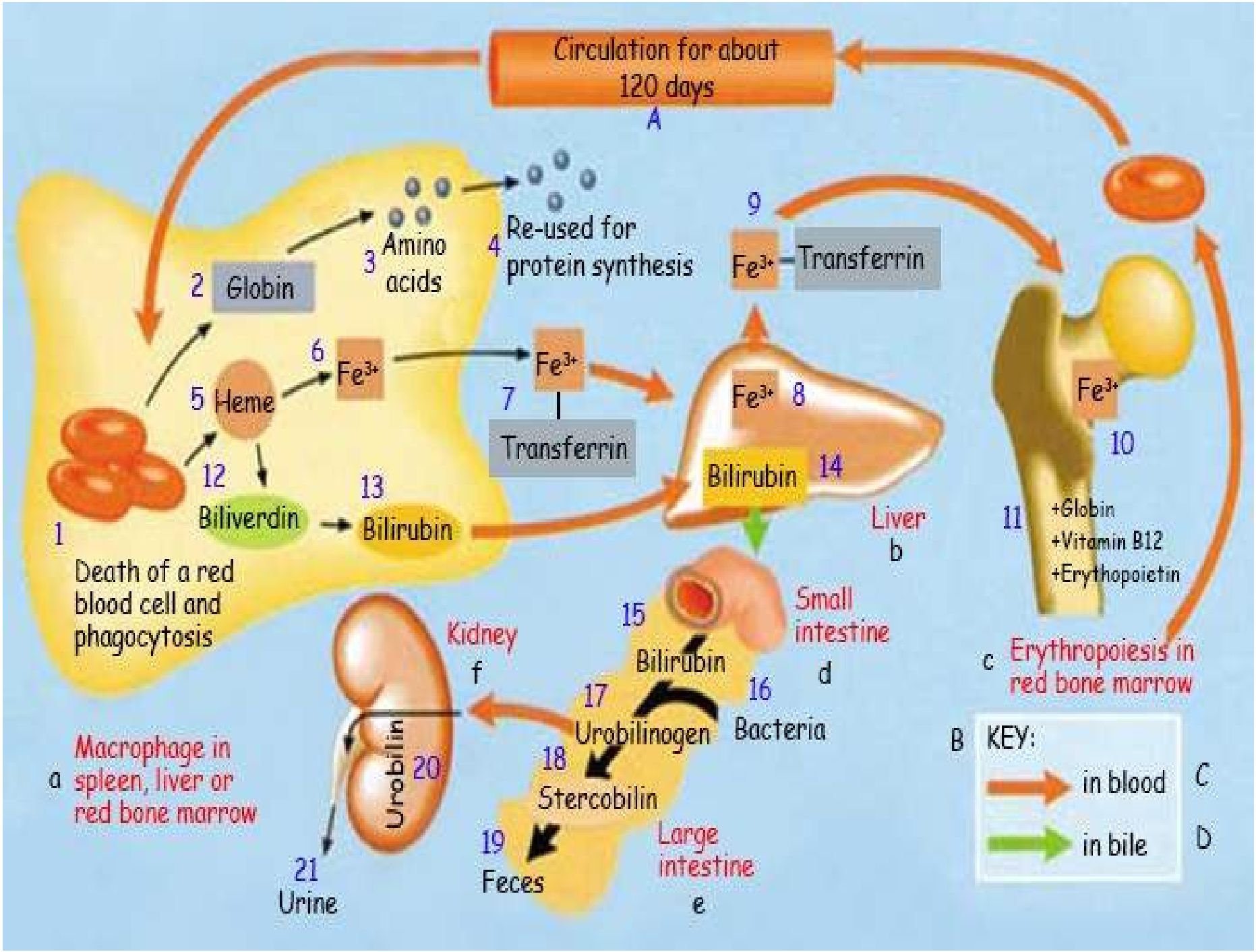
- **Bone marrow erythroid cells** : about **85%** of heme produced in the body. Synthesis rate is **relatively constant** at all times
- **Liver (~10%)**: where **cytochrome P450** is synthesized. Synthesis is up-regulated in response to drug/alcohol metabolism

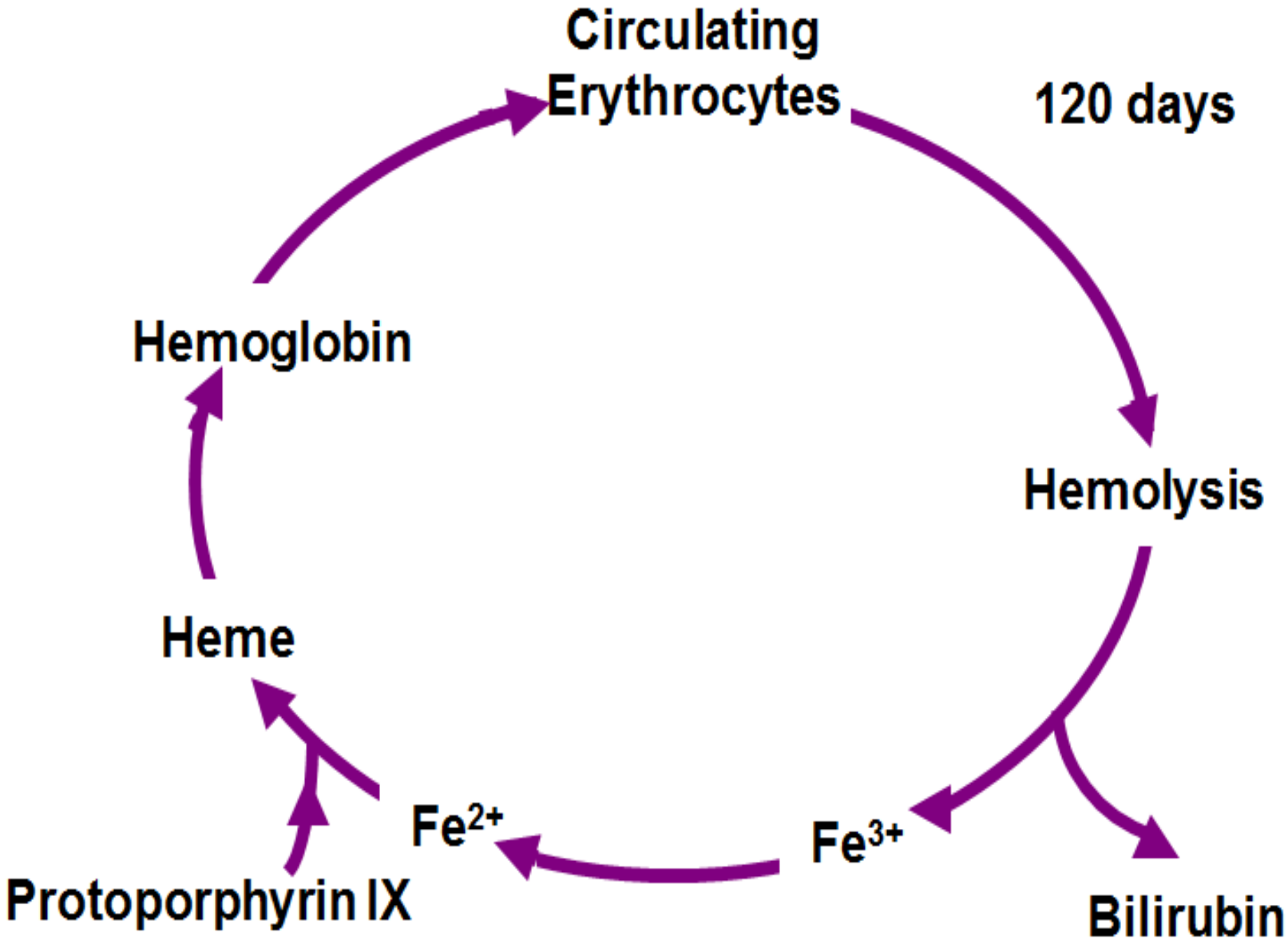
Cellular location:

- **Mitochondria**: the initial reaction and the last three steps
- **Cytosol**: the intermediate 4 steps

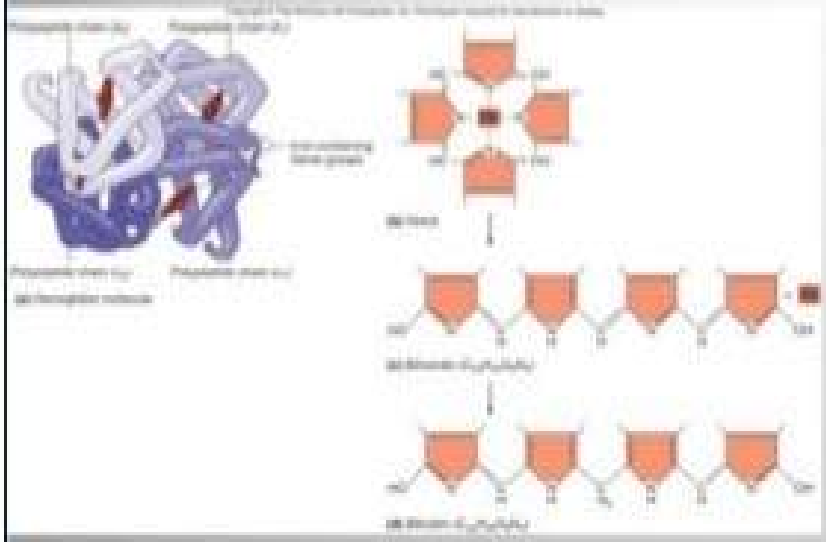
Note: mature RBCs don't have a mitochondria and cannot make heme


Cytosol
Mitochondrion
Ferrochelatase
Heme
Protoporphyrin IX
Fe2+
Coproporphyrinogen III
Uroporphyrinogen decarboxylase
Uroporphyrinogen III
Hydroxymethylbilane
Porphobilinogen deaminase
Porphobilinogen
Pb2+
ALA dehydratase
B6
ALA synthase
Glycine + Succinyl-CoA
δ-aminolevulinic acid


Circulation for about 120 days
A
1
Death of a red blood cell and phagocytosis
2 Globin
3 Amino acids
4 Re-used for protein synthesis
5 Heme
6 Fe3+
7 Fe3+ Transferrin
8 Fe3+
9 Fe3+ Transferrin
10 Fe3+
11 +Globin +Vitamin B12 +Erythopoietin
12 Biliverdin
13 Bilirubin
14 Bilirubin
Liver
b
Small intestine
d
15 Bilirubin
16 Bacteria
17 Urobilinogen
18 Stercobilin
19 Feces
Large intestine
e
Kidney
f
20 Urobilin
21 Urine
a Macrophage in spleen, liver or red bone marrow
c Erythropoiesis in red bone marrow
B KEY:
in blood
in bile
C
D


Circulating Erythrocytes
120 days
Hemolysis
Bilirubin
Fe3+
Fe2+
Protoporphyrin IX
Heme
Hemoglobin

Hemoglobin breakdown



Hemoglobin

Heme + Globin

Amino acids
- recycled by body

Biliverdin + Iron (Fe^{2+})
- Reused by liver in synthesis of new hemoglobin

Bilirubin
- **Excreted as bile pigment**

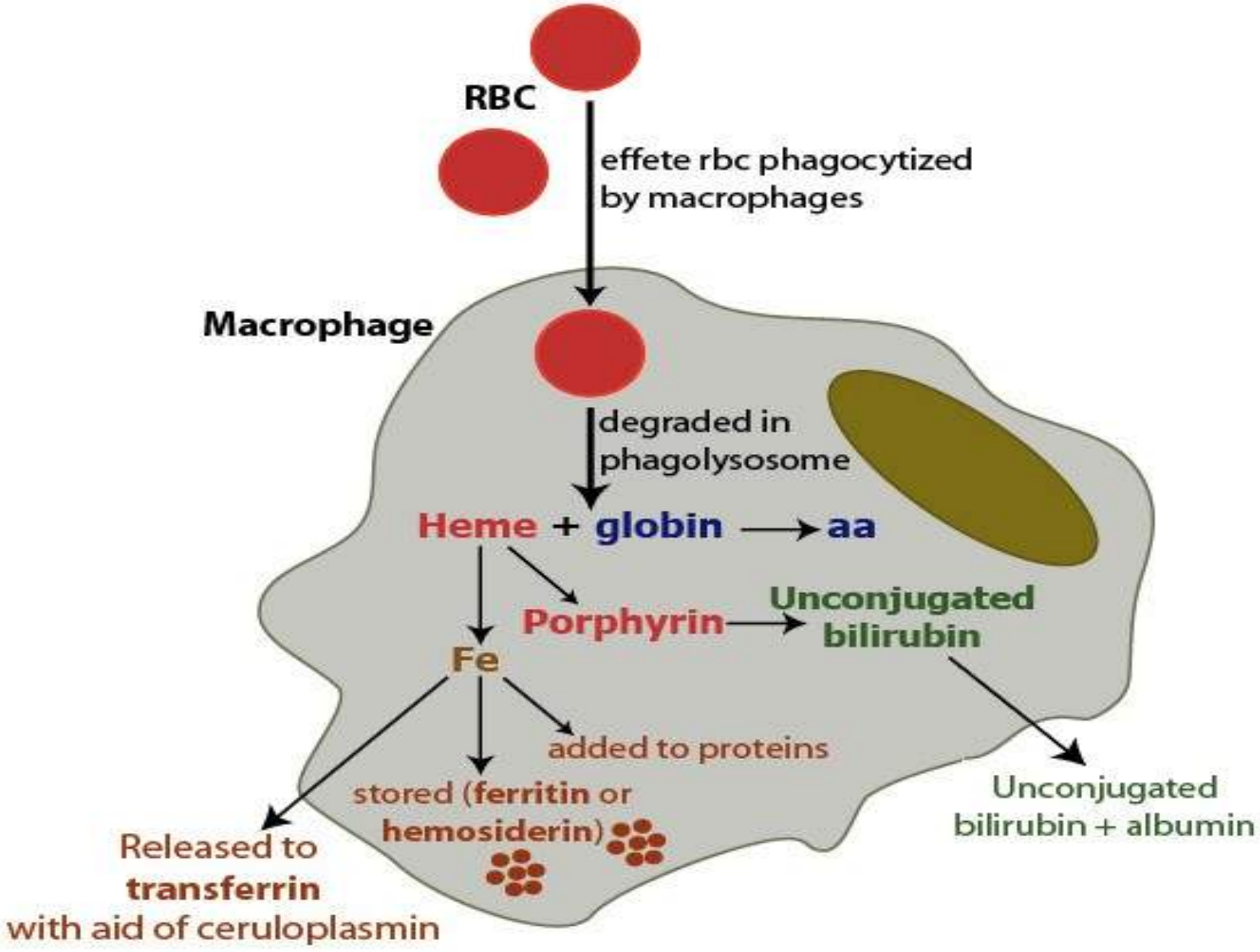
Jaundice: yellowish staining of skin caused by the accumulation of bile pigments.

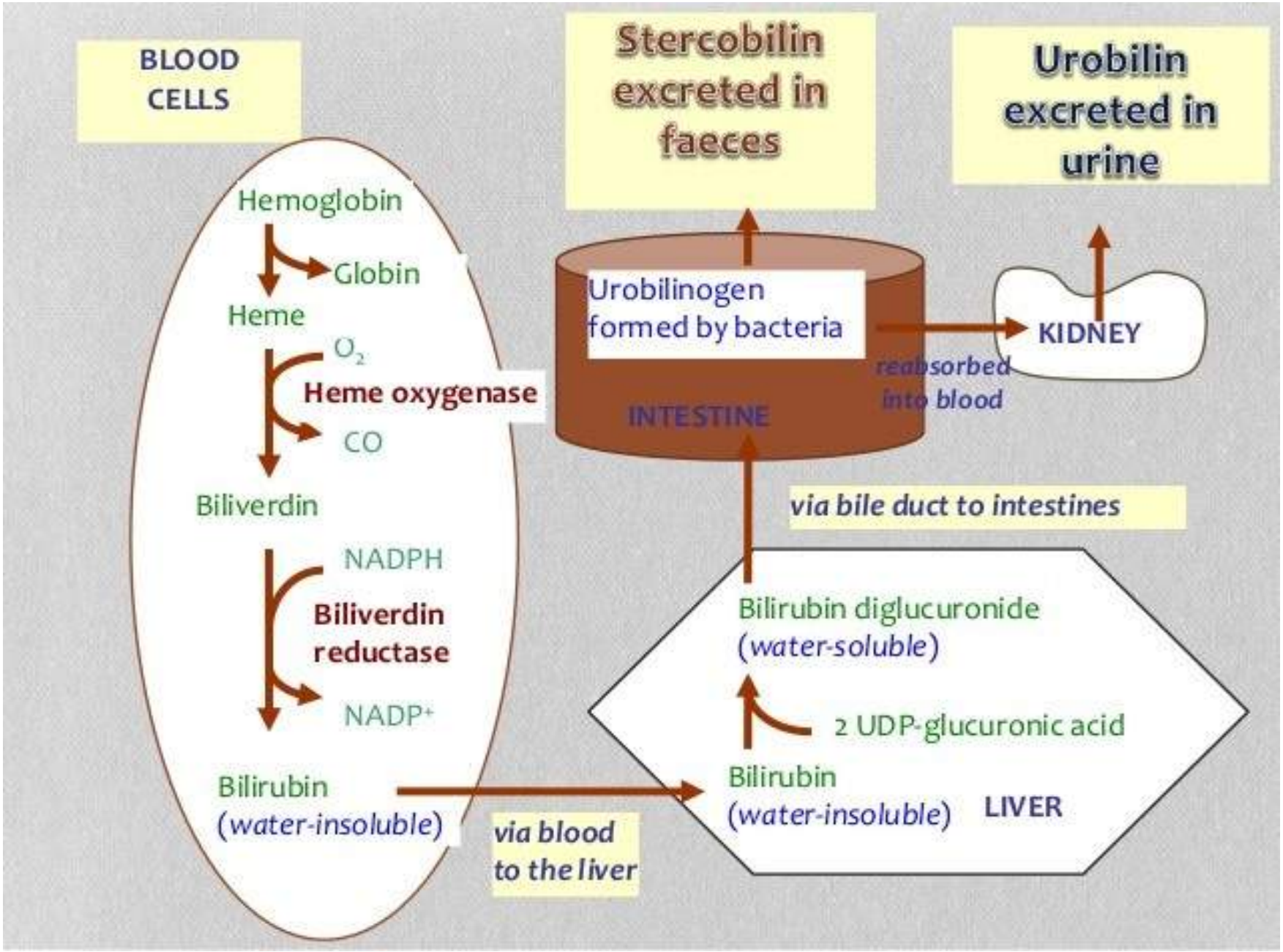
Jaundice may indicate increased RBC destruction, destruction of liver tissue, or blockage of the biliary tract.

Degradation of Heme

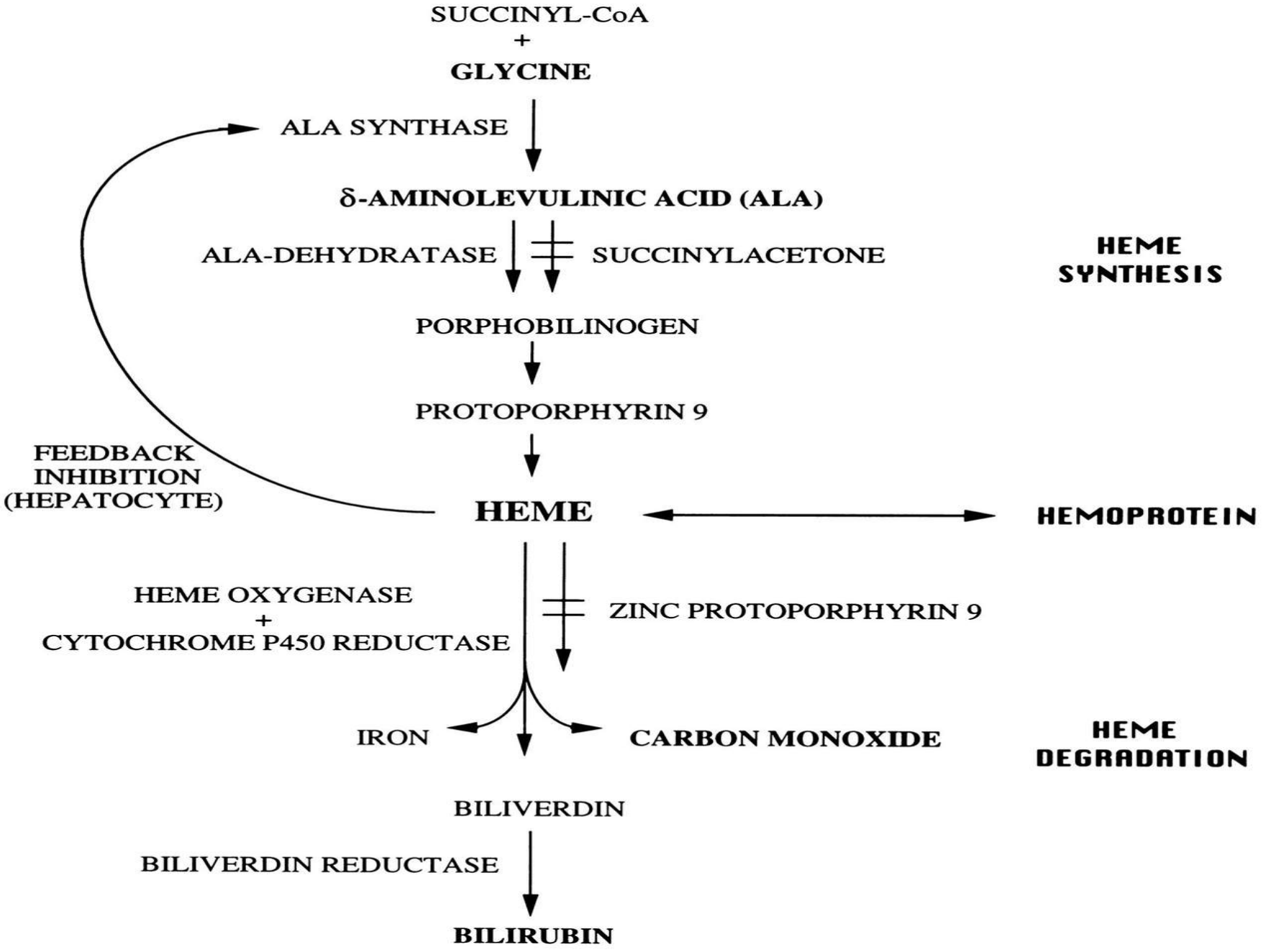
- After approximately 120 days in the circulation, red blood cells are taken up and degraded by the reticulo-endothelial system, particularly in the liver and spleen.
- Approximately 85% of heme destined for degradation comes from senescent red blood cells, and 15% is from turnover of immature red blood cells and cytochromes from nonerythroid tissues.

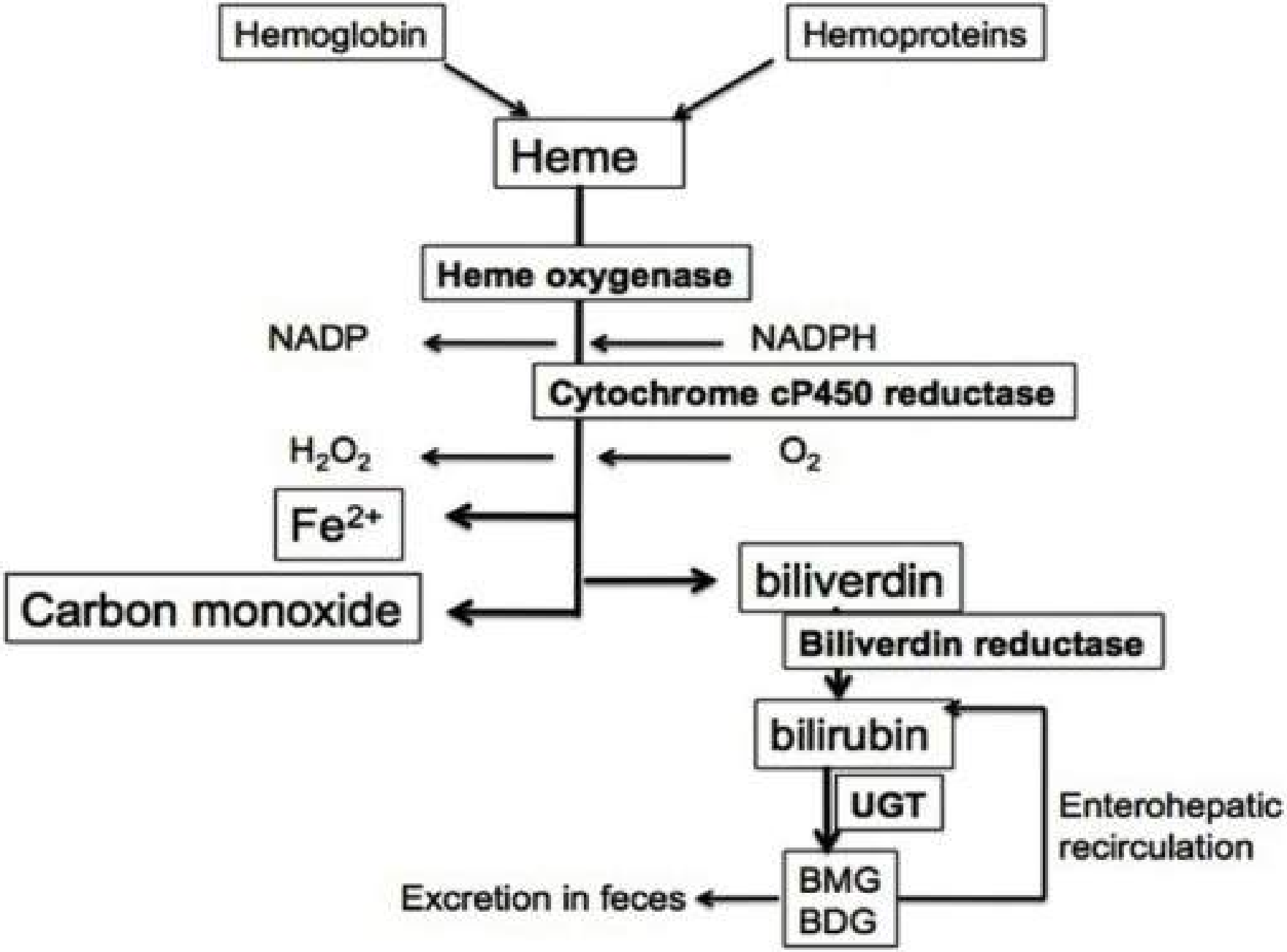
- Physiological conditions: 1-2x 10^8 erythrocytes are destroyed per hour
- A 70 kg human turns over 6g of hemoglobin
- Hemoglobin is destroyed, its globin is degraded to its constituent amino acids and reused
- The **iron** of heme enters the iron pool, also for reuse.
- The iron-free **porphyrin** portion of heme is also degraded, mainly in the reticuloendothelial cells of the liver, spleen, and bone marrow.

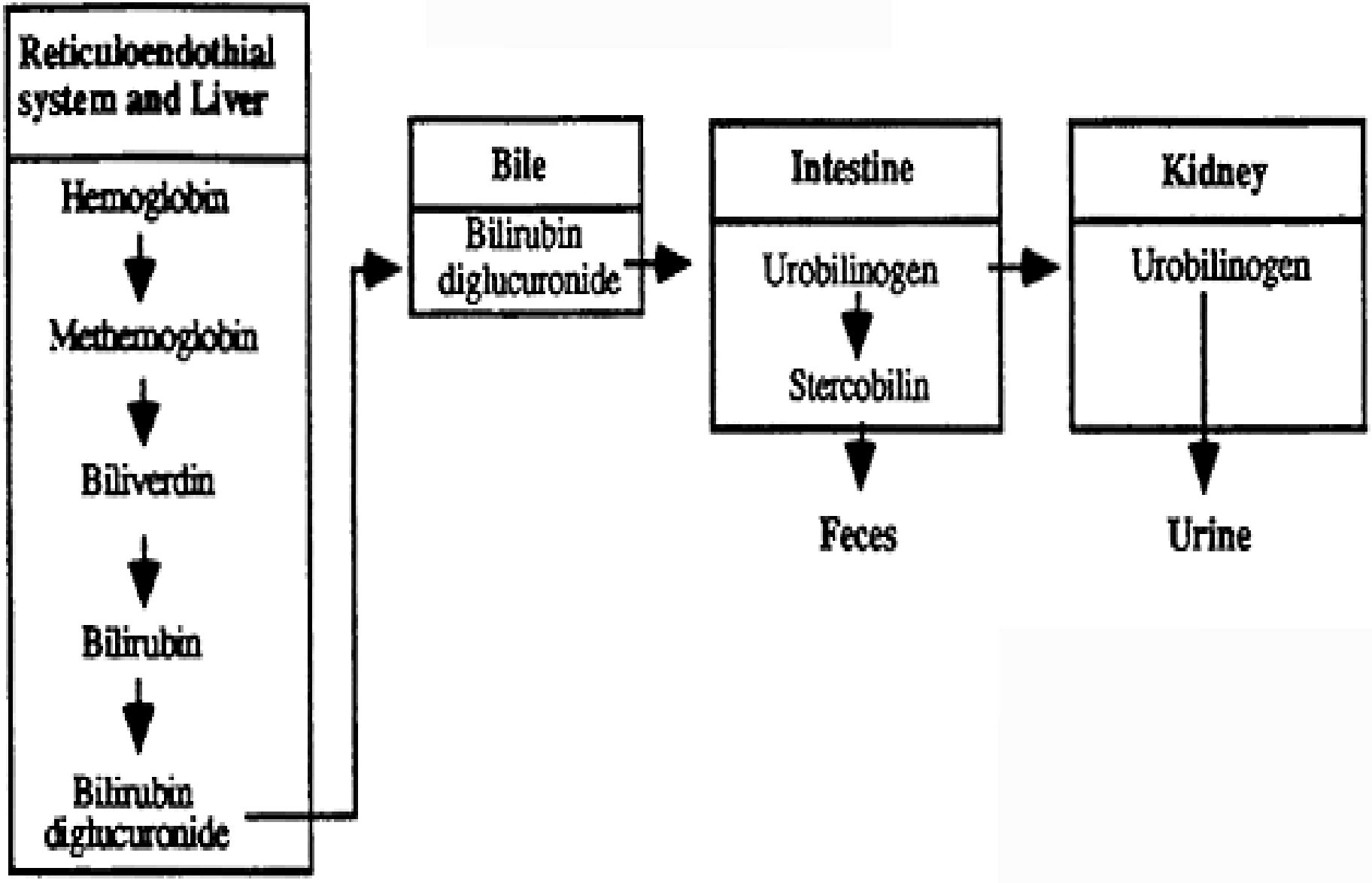

RBC
effete rbc phagocytized by macrophages
Macrophage
degraded in phagolysosome
Heme + globin → aa
Porphyrin
Unconjugated bilirubin
Fe
added to proteins
stored (ferritin or hemosiderin)
Released to transferrin with aid of ceruloplasmin
Unconjugated bilirubin + albumin


BLOOD CELLS
Stercobilin excreted in faeces
Urobilin excreted in urine
Hemoglobin
Globin
Heme
O_2
Heme oxygenase
CO
Biliverdin
NADPH
Biliverdin reductase
$NADP^+$
Bilirubin (water-insoluble)
via blood to the liver
Urobilinogen formed by bacteria
INTESTINE
reabsorbed into blood
KIDNEY
via bile duct to intestines
Bilirubin diglucuronide (water-soluble)
2 UDP-glucuronic acid
Bilirubin (water-insoluble)
LIVER

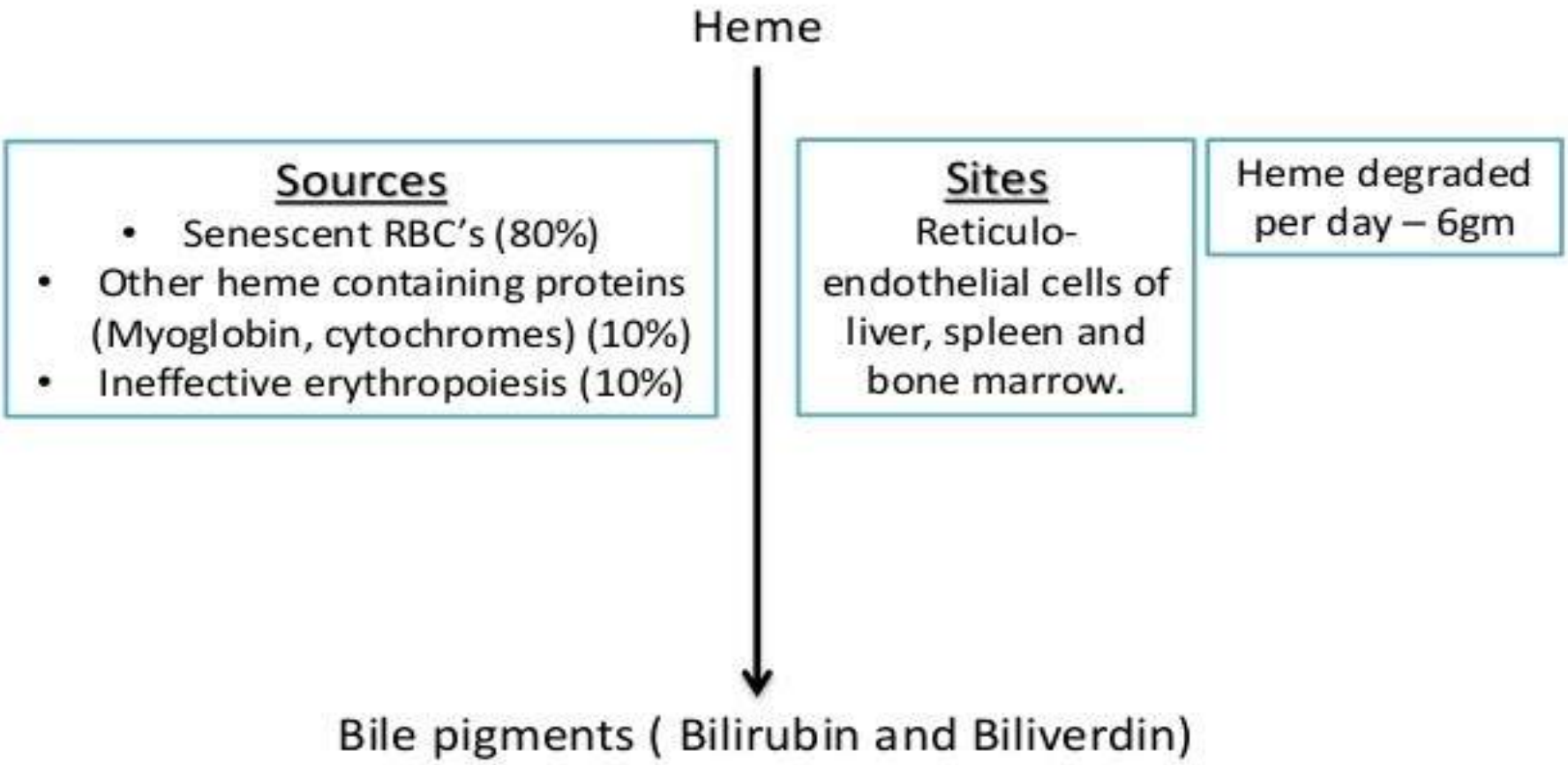
- The catabolism of heme from all of the heme proteins appears to be carried out in the microsomal fractions of cells by a complex enzyme system called heme oxygenase.
- By the time the heme derived from heme proteins reaches the oxygenase system, the iron has usually been oxidized to the ferric form, constituting **hemin**.


SUCCINYL-CoA
+
GLYCINE
ALA SYNTHASE
δ-AMINOLEVULINIC ACID (ALA)
ALA-DEHYDRATASE
SUCCINYLACETONE
HEME SYNTHESIS
PORPHOBILINOGEN
PROTOPORPHYRIN 9
FEEDBACK INHIBITION (HEPATOCYTE)
HEME
HEMOPROTEIN
HEME OXYGENASE
+
CYTOCHROME P450 REDUCTASE
ZINC PROTOPORPHYRIN 9
IRON
CARBON MONOXIDE
HEME DEGRADATION
BILIVERDIN
BILIVERDIN REDUCTASE
BILIRUBIN


Hemoglobin
Hemoproteins
Heme
Heme oxygenase
NADP
NADPH
Cytochrome cP450 reductase
H_2O_2
O_2
Fe^{2+}
Carbon monoxide
biliverdin
Biliverdin reductase
bilirubin
UGT
Enterohepatic recirculation
Excretion in feces
BMG
BDG


Reticuloendothial system and Liver
Hemoglobin
Methemoglobin
Biliverdin
Bilirubin
Bilirubin diglucuronide
Bile
Bilirubin diglucuronide
Intestine
Urobilinogen
Stercobilin
Feces
Kidney
Urobilinogen
Urine

Catabolism of heme



Formation of Bilirubin from Heme

Heme is degraded **in RE system (esp. liver & spleen)**
85% from RBCs
15% from turnover of immature RBCs & cytochromes

Heme

⇩ heme oxygenase

biliverdin (green)

⇩

bilirubin (red-orange)
bile pigments

⇩

In Blood with albumin
UNCONJUGATED BILIRUBIN
(or INDIRECT BILITUBIN)

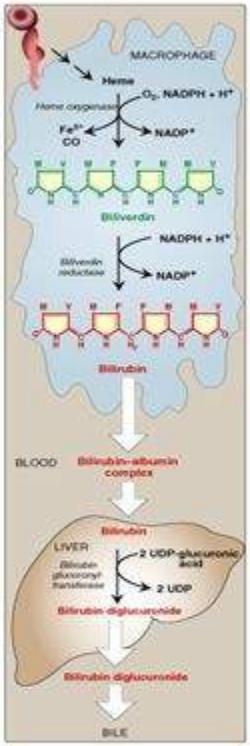


Figure 21.8
Formation of bilirubin from heme.

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Salicylates & sulfonamides can displace bilirubin from albumin & so bilirubin enters CNS causing **neural damage**

Degradation of Heme to bile pigments

Biliverdin reductase

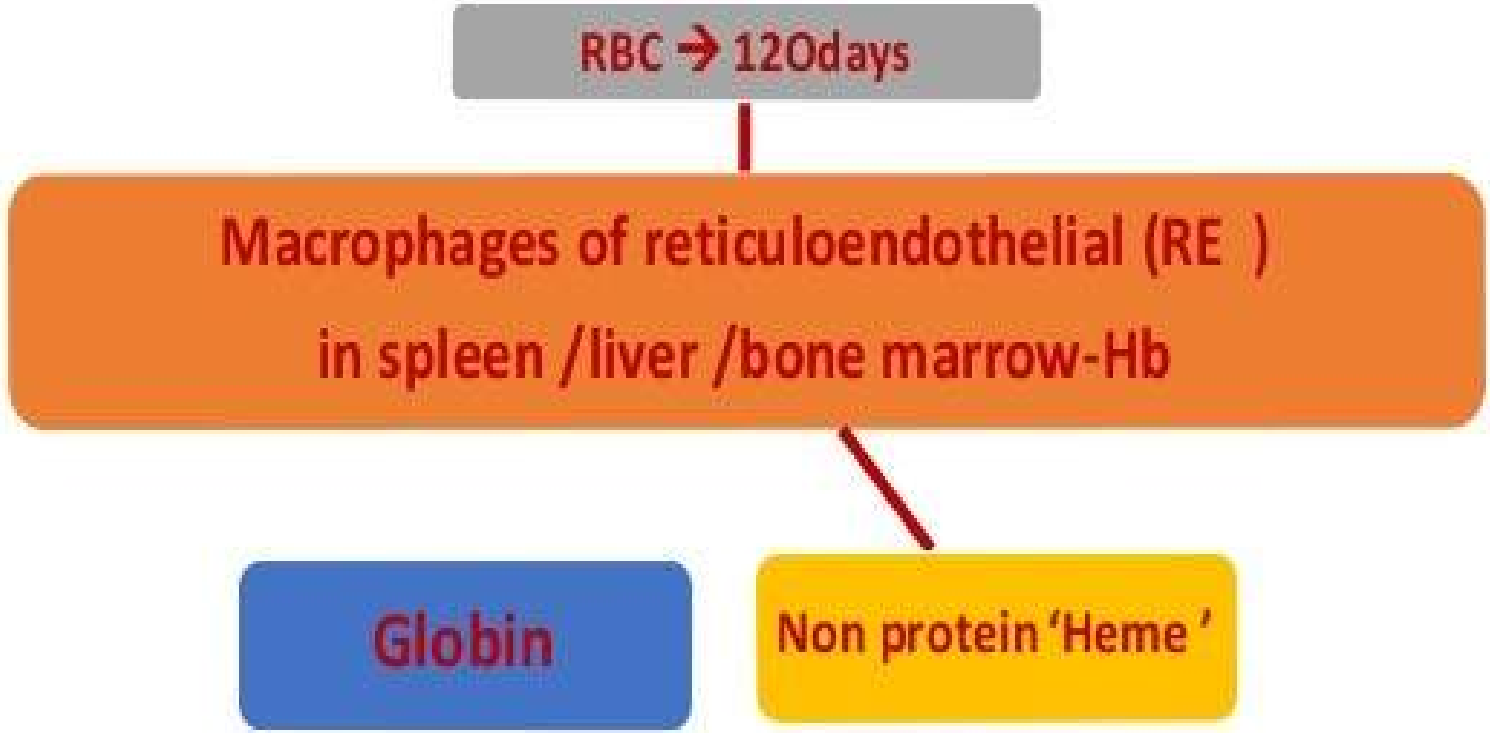
Biliverdin (green) → **Bilirubin** (non functional therefore excreted)

1gm hemoglobin * → 35 mg Bilirubin

Biliverdin + Bilirubin → Heme derivatives

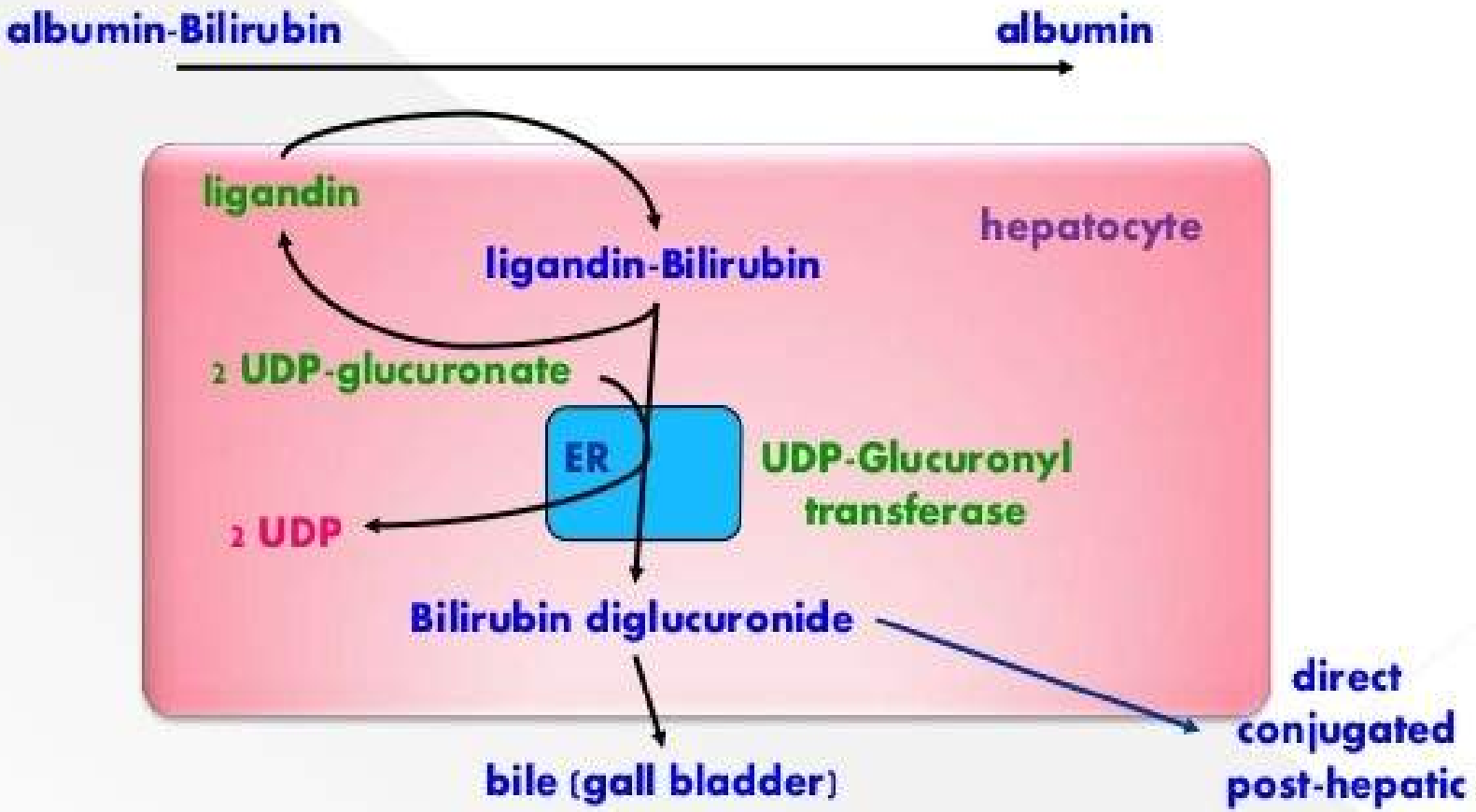
- (* sources- 80% RBC + 10% ineffective erythropoiesis + 10% Myoglobin)
- **6 gm /day heme broken down /resynthesis → Bilirubin (300 mg /day) = 250 mg/day Hb + 50 mg/day myoglobin**

Degradation of Heme to bile pigments

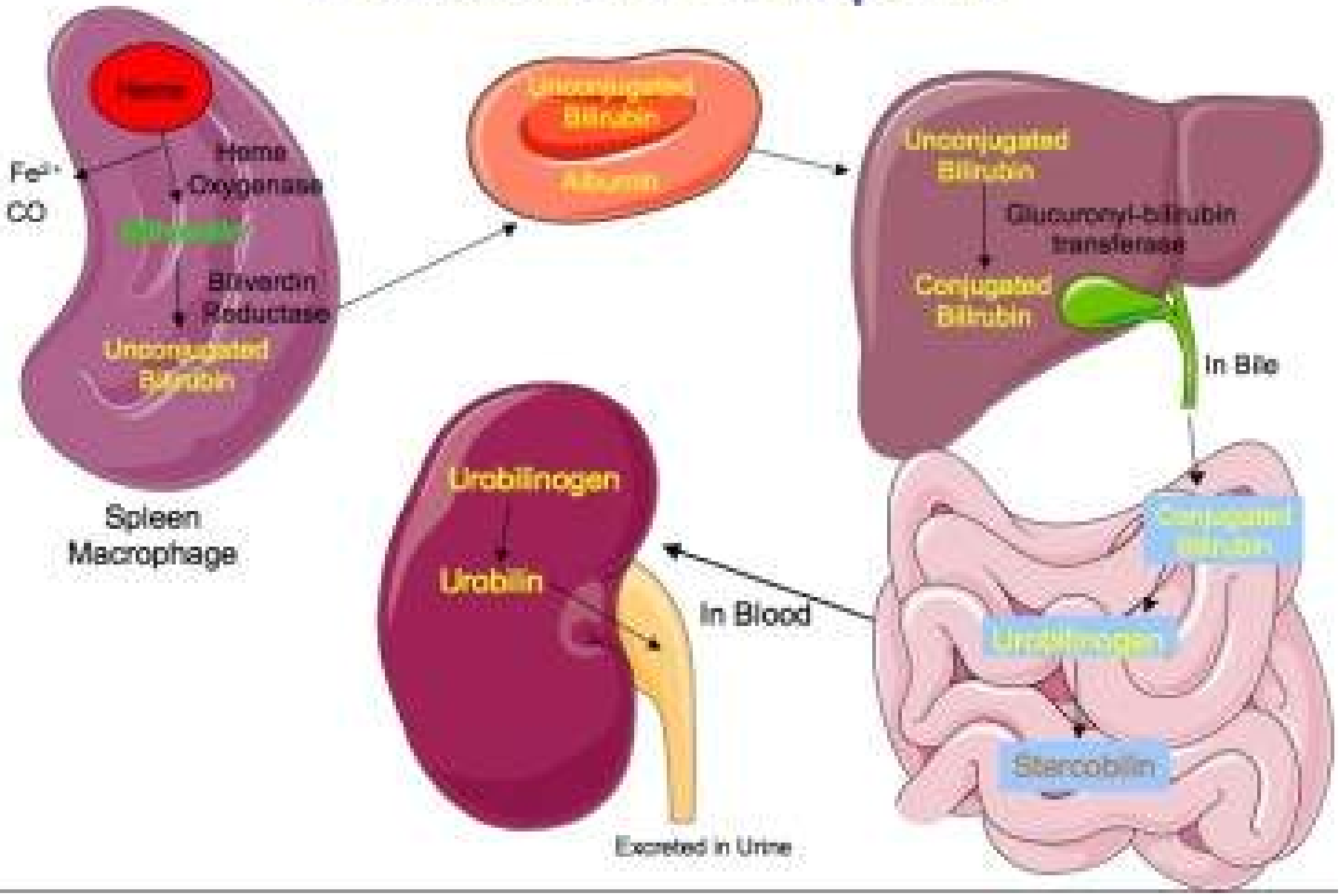


6 gm /day heme broken down /resynthesis→ Bilirubin (300 mg /day) = 250 mg/day Hb + 50 mg/day myoglobin

Bilirubin Processing



Bilirubin Transport



Pigment

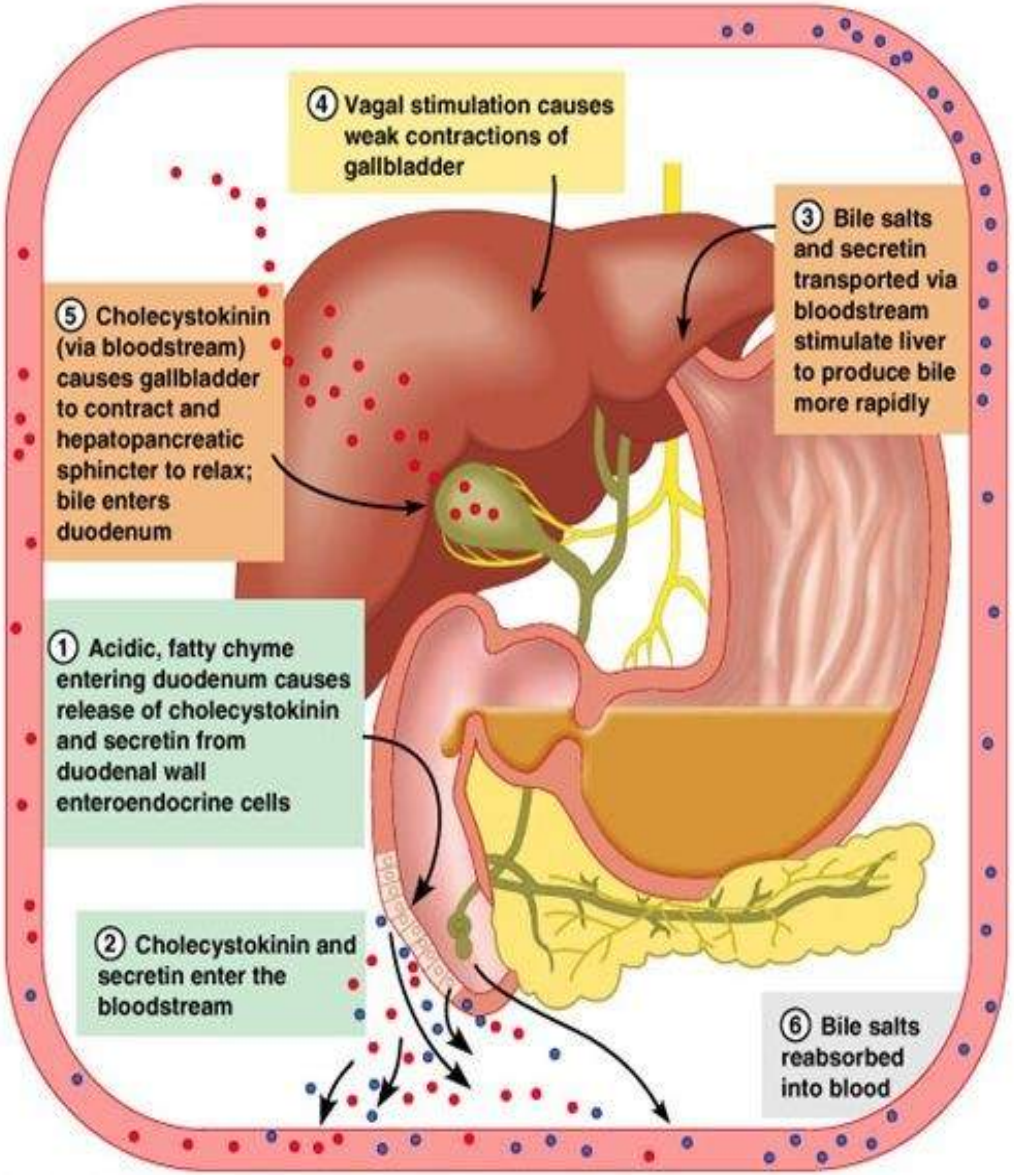
- Bile pigments are endogenous(formed inside the body), coloured compounds – breakdown products of the blood pigment haemoglobin – that are excreted in bile

- The two most important bile pigments are-
  - *Bilirubin*, which is orange or yellow, and
  - It's oxidized form *Biliverdin*, which is green.

Bile Juice

- Bile is a bitter-tasting, dark green to yellowish brown fluid, produced by the liver , it is stored in the gallbladder and upon eating is discharged into the duodenum. .
- The principal function of the gallbladder is to serve as a storage reservoir for bile.
- The main components of bile are water, bile salts, bile pigments, and cholesterol
- Bile salts act as emulsifying agents in the digestion and absorption of fats. Cholesterol and bile pigments from the breakdown of hemoglobin are excreted from the body in the bile.

COMPOSITION of BILE



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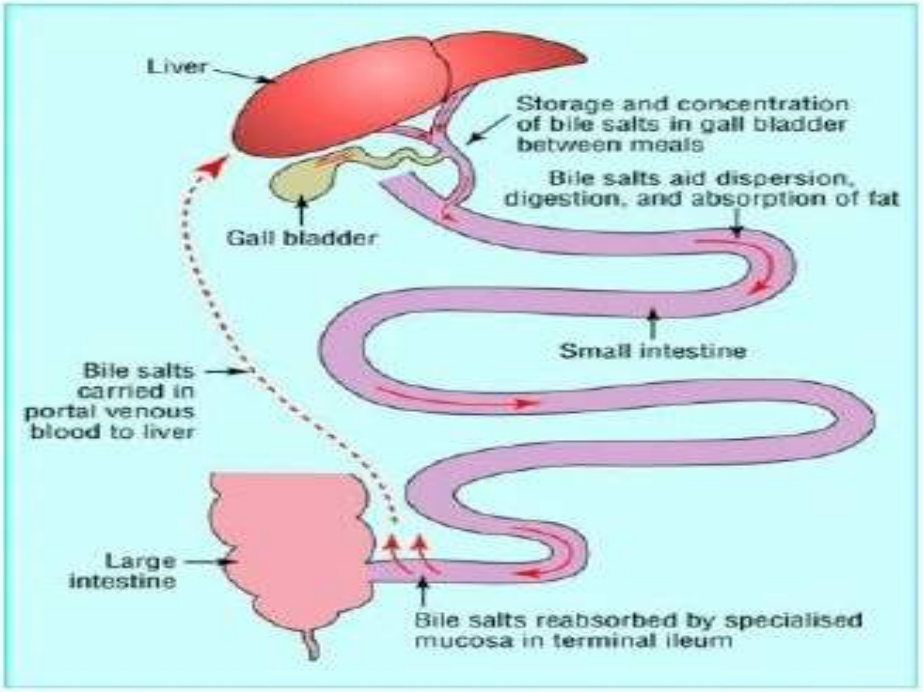
- **Chief bile pigment is bilirubin**, a waste product of the heme of hemoglobin formed during the breakdown of worn-out erythrocytes
  - Absorbed by the liver cells, excreted into bile, and metabolized in the small intestine by resident bacteria
    - One of the breakdown products, urobilinogen, gives feces a brown color
      - In the absence of bile, feces are gray-white in color and have fatty streaks (because essentially no fats are digested or absorbed)
- Stimulus of bile secretion:
  - Bile salts
  - Secretin (released by the intestinal cells)

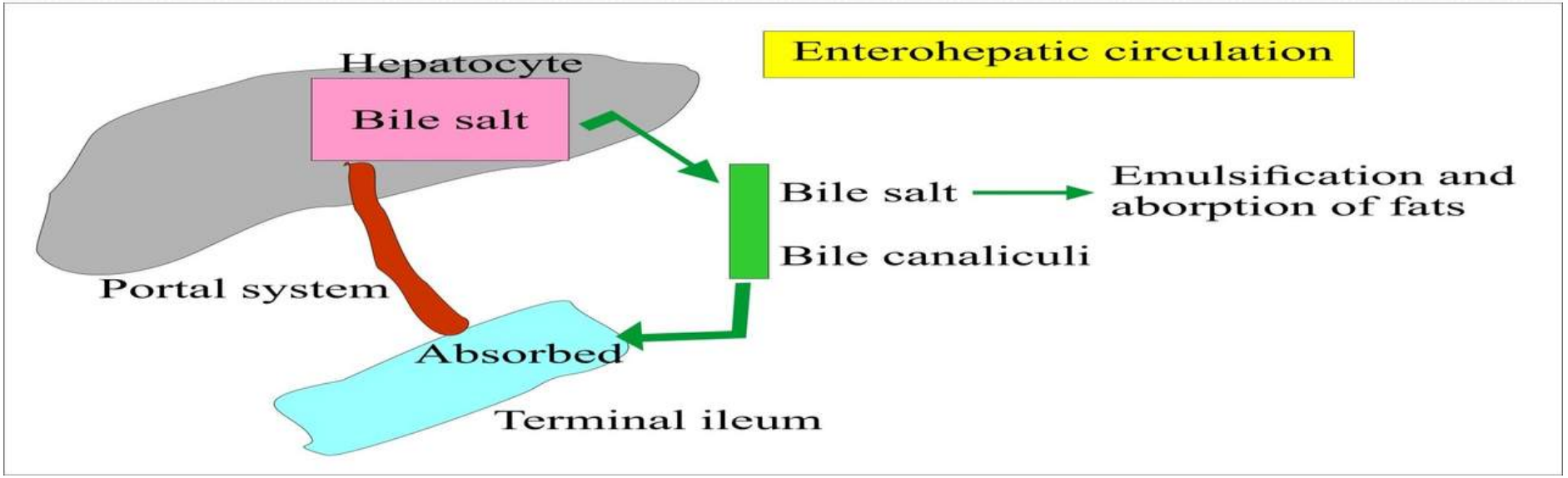
Functions of Liver

- **Bile production**
  - Bile comprises of **bile salts** and **bile pigments** that is stored in the gall bladder till needed in fat digestion
  - **Bile salts** are synthesized from **cholesterol**
  - **Bile pigments** (yellowish-green) are from the incorporation of the **by-products of red blood cell** disassembly
  - **Gallstones** are an accumulation of cholesterol crystals that can cause blockage of the bile/biliary duct and increase pressure of the gall bladder.
  - Accumulation stems from bile constituents' imbalance.

Enterohepatic Circulation

- Bile acids are stored in the gallbladder
- During digestion bile acids are absorbed through the gut mostly uncongugated (some congugated)
- In the terminal ileum the bile salts are reabsorbed and carried through the portal blood circulation to be reconjugated and secreted back into bile



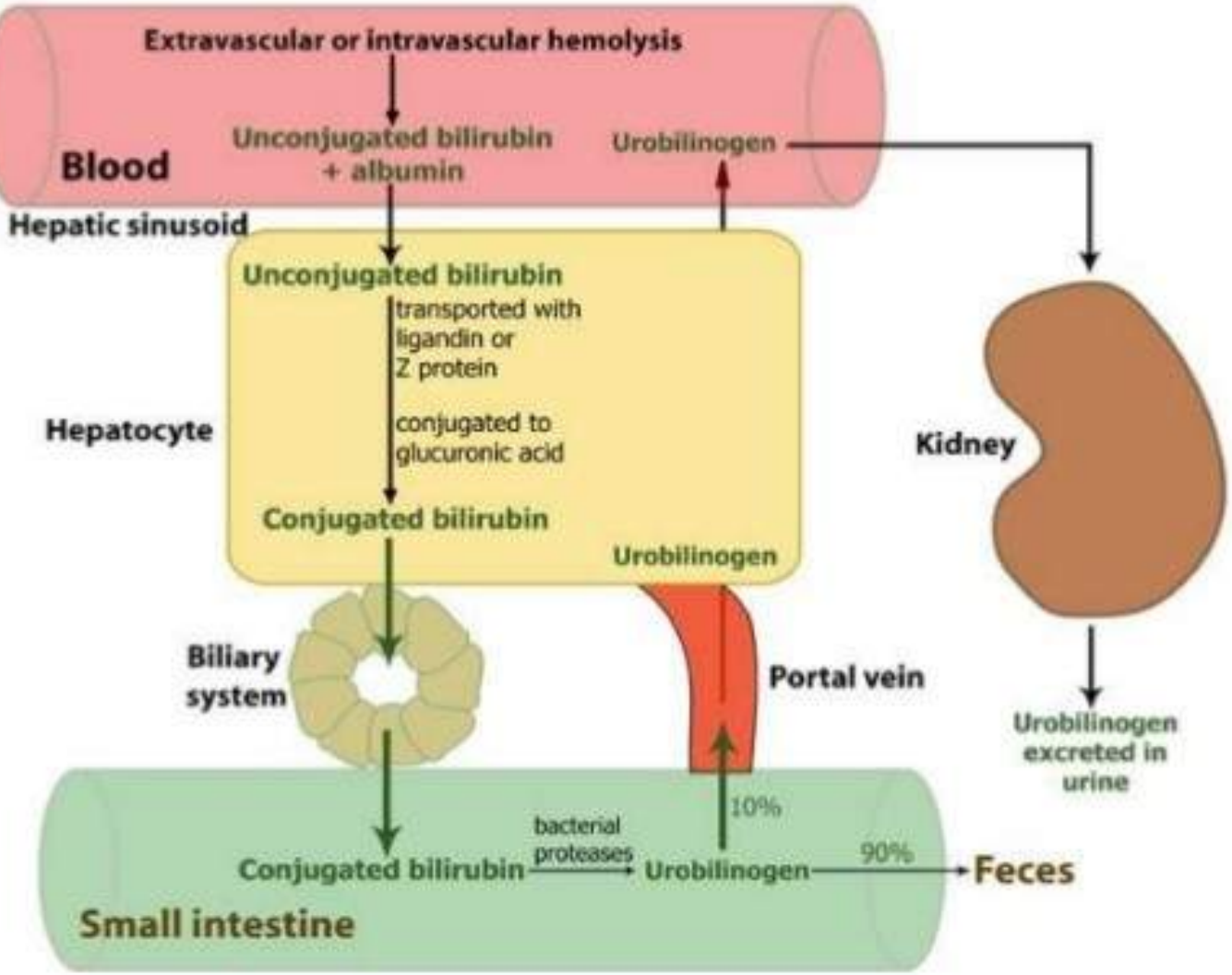


Disorders of bile pigment metabolism

Most liver diseases and several nonhepatic disorders are companied by jaundice.

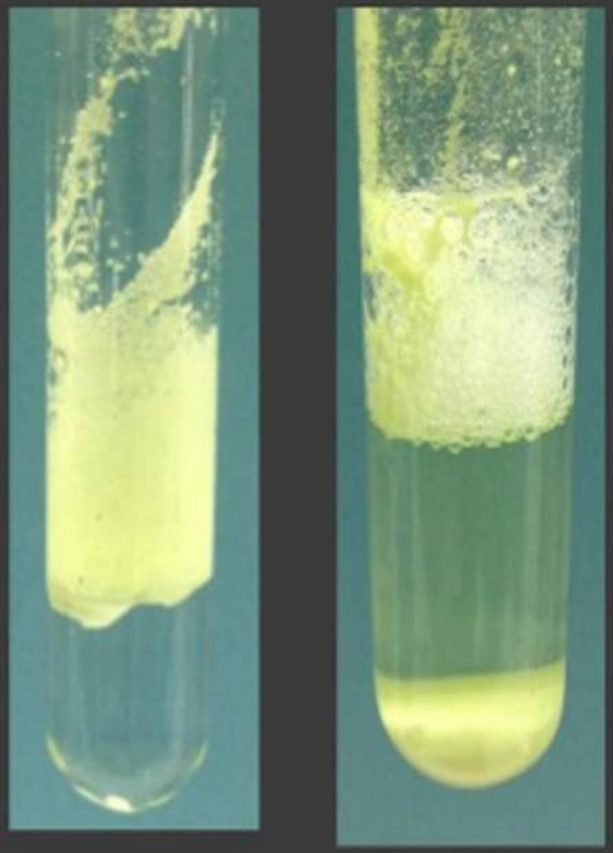
1) **A excessive load of bilirubin** (hemolytic diseases , chronic hemolytic anemia)
2) **Defective transport into the hepatocyte**. This transport defect is known as Gilberts disease.
3) **Impairment in esterification of bilirubin**.
   A) physiologic jaundice of new born due to the UDPG transferase enzyme is not fully developed.
   B)Hemolytic disease of new born due to Rh and ABO system incompatibility.
4) **Congenital deficiency of UDPG transferase**.
   Known as the Crigler-Najjar syndrome .

Excretion of bilirubin



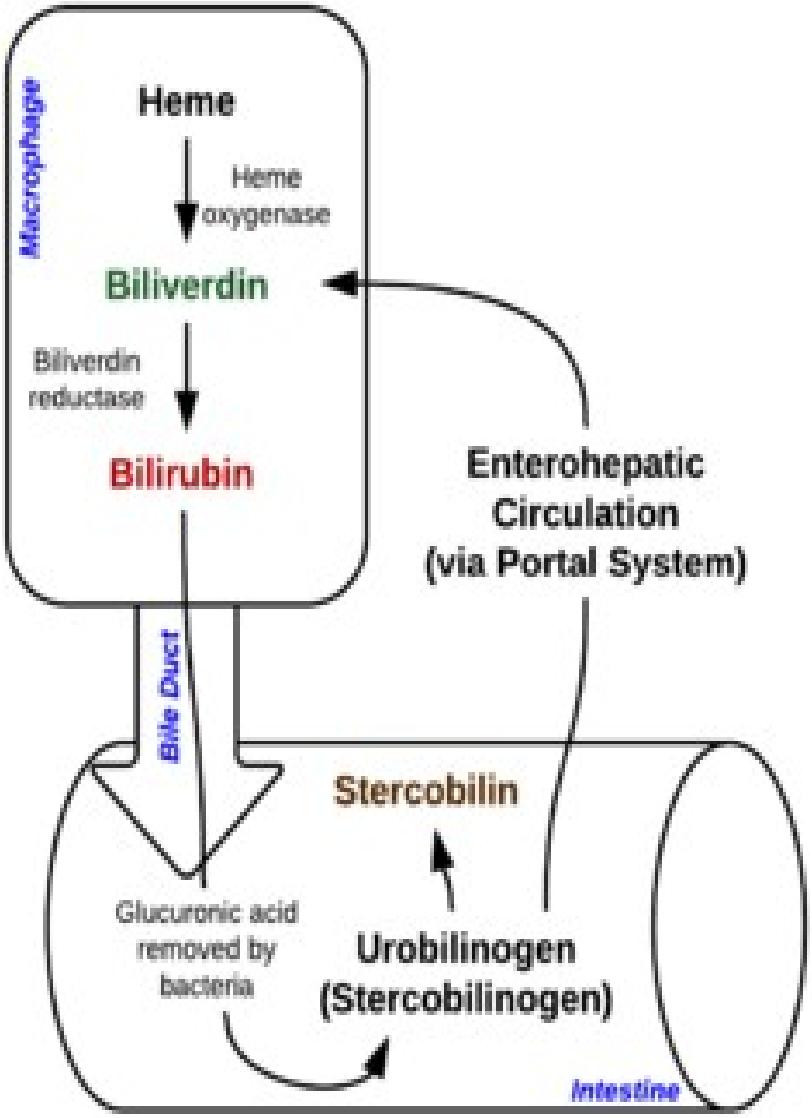
Hay's test for bile salts

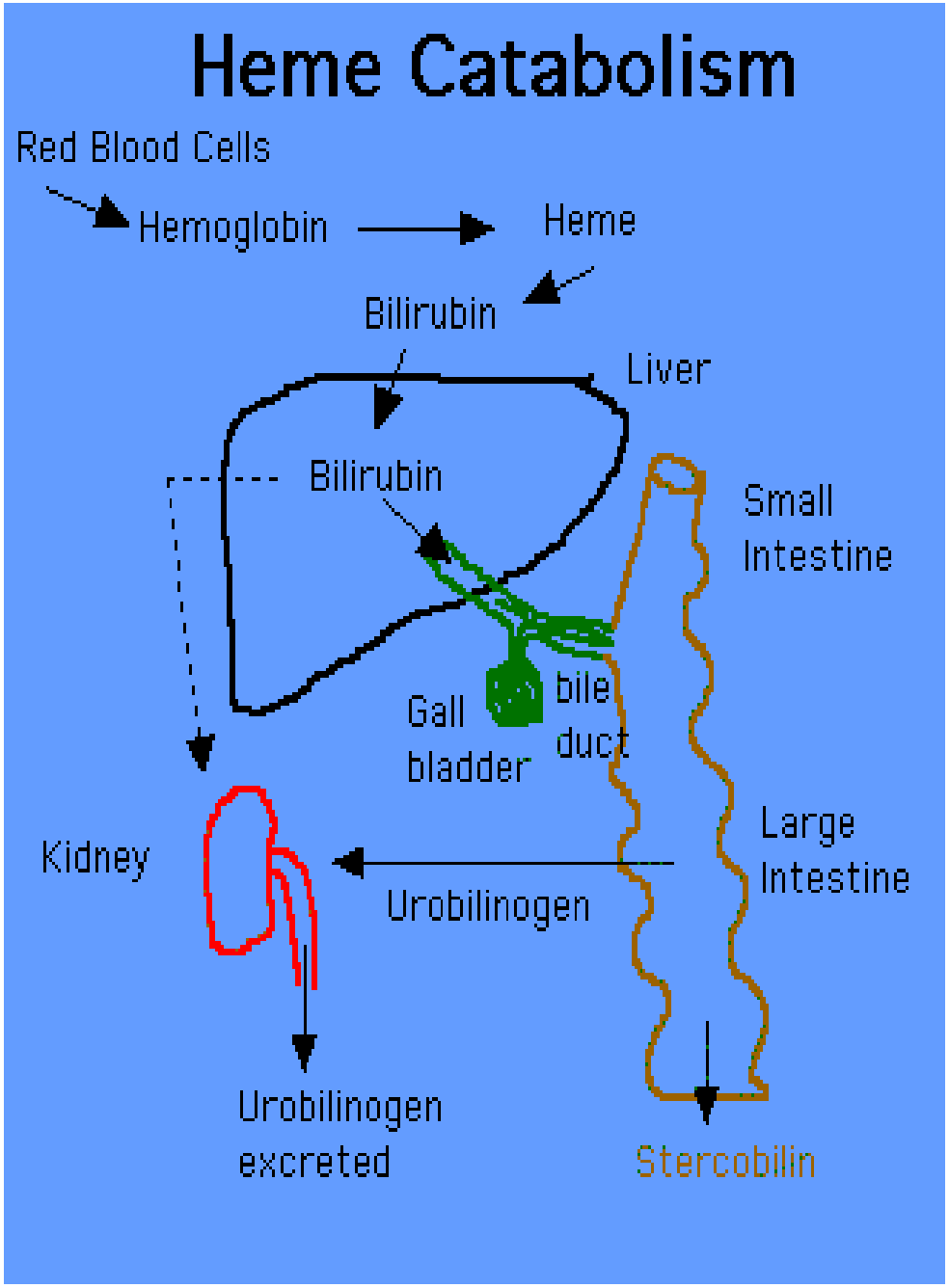
Its principle, reagent and procedure

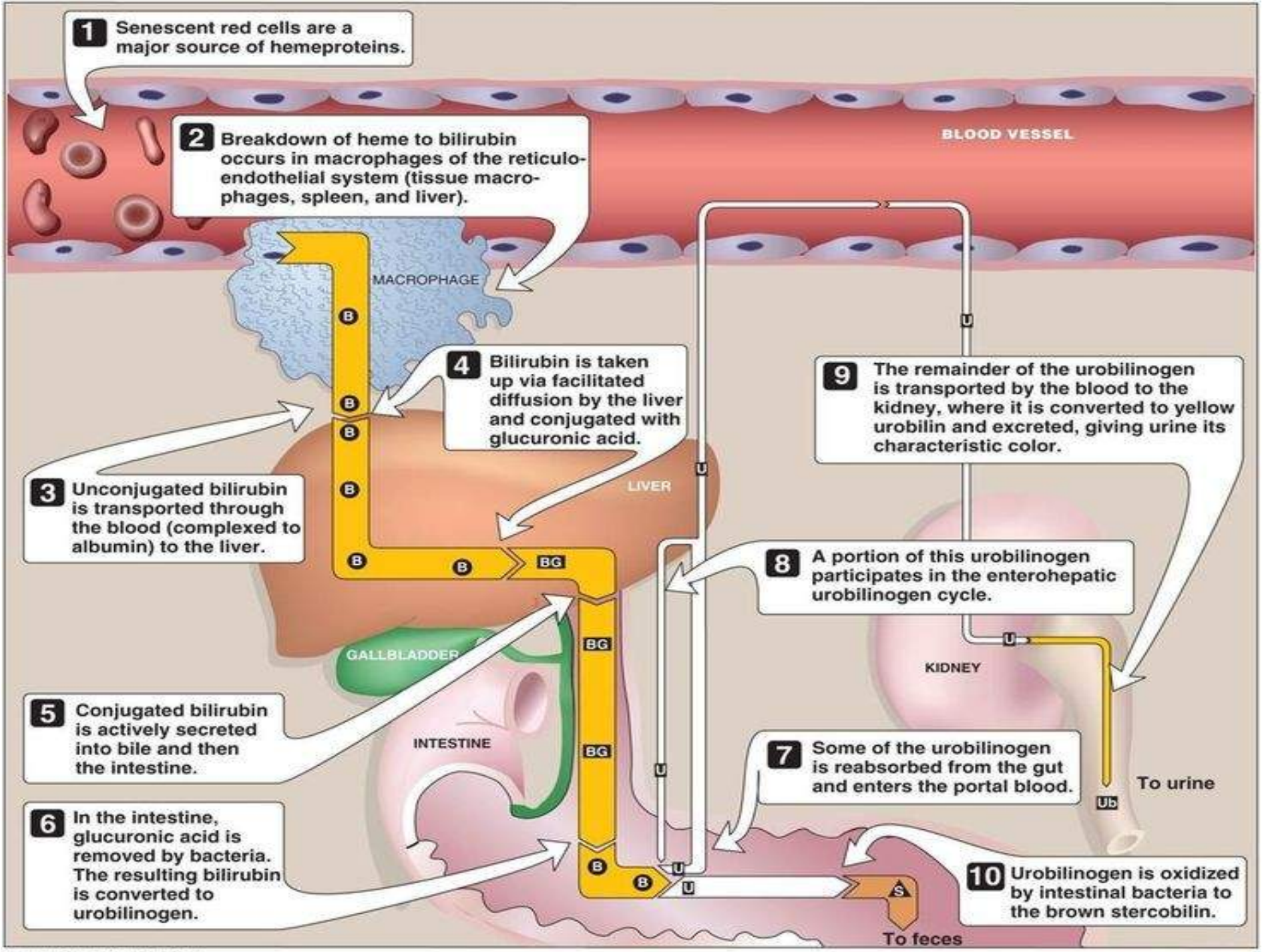


Control for comparison

Positive test

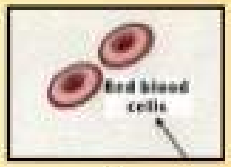

Macrophage
Heme
Heme oxygenase
Biliverdin
Biliverdin reductase
Bilirubin
Enterohepatic Circulation (via Portal System)
Bile Duct
Stercobilin
Glucuronic acid removed by bacteria
Urobilinogen (Stercobilinogen)
Intestine


Heme Catabolism
Red Blood Cells
Hemoglobin
Heme
Bilirubin
Liver
Bilirubin
Small Intestine
Gall bladder
bile duct
Kidney
Large Intestine
Urobilinogen
Urobilinogen excreted
Stercobilin


1 Senescent red cells are a major source of hemeproteins.
2 Breakdown of heme to bilirubin occurs in macrophages of the reticulo-endothelial system (tissue macrophages, spleen, and liver).
BLOOD VESSEL
MACROPHAGE
B
4 Bilirubin is taken up via facilitated diffusion by the liver and conjugated with glucuronic acid.
3 Unconjugated bilirubin is transported through the blood (complexed to albumin) to the liver.
LIVER
BG
9 The remainder of the urobilinogen is transported by the blood to the kidney, where it is converted to yellow urobilin and excreted, giving urine its characteristic color.
U
8 A portion of this urobilinogen participates in the enterohepatic urobilinogen cycle.
GALLBLADDER
KIDNEY
Ub
5 Conjugated bilirubin is actively secreted into bile and then the intestine.
INTESTINE
7 Some of the urobilinogen is reabsorbed from the gut and enters the portal blood.
To urine
6 In the intestine, glucuronic acid is removed by bacteria. The resulting bilirubin is converted to urobilinogen.
S
10 Urobilinogen is oxidized by intestinal bacteria to the brown stercobilin.
To feces

Extravascular Pathway for RBC Destruction

(Liver, Bone marrow, & Spleen)



Phagocytosis & Lysis

Hemoglobin

Globin → Amino acids → Amino acid pool

Heme → Fe^{2+} → Recycled

Heme → Bilirubin → Excreted

Thank you!!